Problem 16.9

You will retire in 30 years. At the beginning of each month until you retire, you will invest X earning interest at 9% convertible monthly. Starting at year 30, you will withdraw \$4,000 at the beginning of each month for the next 15 years. Also, starting at year 30, your fund will only earn interest at 6% convertible monthly. Find X such that your account will be empty after the last withdrawal.

Solution

During the accumulation period, the nominal annual interest rate convertible monthly is 9%, so the monthly rate is

$$j_1 = \frac{0.09}{12} = 0.0075.$$

There are

$$30(12) = 360$$

monthly deposits made at the beginning of each month, so their accumulated value at retirement is

$$X\ddot{s}_{\overline{360}|}.$$

During the withdrawal period, the nominal annual interest rate convertible monthly is 6%, so the monthly rate is

$$j_2 = \frac{0.06}{12} = 0.005.$$

There are

$$15(12) = 180$$

monthly withdrawals of \$4,000 made at the beginning of each month. The value needed at retirement is

$$4000\ddot{a}_{\overline{180}|}.$$

Equating the accumulated value of the deposits to the present value of the withdrawals at retirement gives

$$X\ddot{s}_{\overline{360}|} = 4000\ddot{a}_{\overline{180}|}.$$

Thus,

$$X = \frac{4000\ddot{a}_{\overline{180}|}}{\ddot{s}_{\overline{360}|}}.$$

Using

$$\ddot{s}_{\overline{n}|} = (1+j)s_{\overline{n}|} \quad \text{and} \quad \ddot{a}_{\overline{n}|} = (1+j)a_{\overline{n}|},$$

we get

$$
\begin{aligned}
X &= \frac{4000(1.005)\left(\dfrac{1-(1.005)^{-180}}{0.005}\right)}{(1.0075)\left(\dfrac{(1.0075)^{360}-1}{0.0075}\right)} \\
&\approx 258.2764.
\end{aligned}
$$

Hence, the required monthly investment is

$$\boxed{X \approx \$258.28}.$$